

3. COURSE OUTCOMES AND PROGRAM OUTCOMES: (120)

Department of Civil Engineering have twelve Program outcomes as defined by NBA and four Program Specific Outcomes.

3.1. Establish the correlation between the courses and the Program Outcomes (POs) and Program Specific Outcomes (PSOs) (20)

(Program Outcomes as mentioned in Annexure I and Program Specific Outcomes as defined by the Program)

PROGRAM OUTCOMES (POs)

- 1. Engineering knowledge:** Apply the knowledge of mathematics, science, electrical engineering fundamentals and specialization to the solution of complex electrical engineering problems.
- 2. Problem analysis:** Identify, formulate, review research literature, and analyze complex electrical engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- 3. Design/development of solutions:** Design solutions for complex electrical engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- 4. Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- 5. Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling of complex engineering activities with an understanding of the limitations.
- 6. The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- 7. Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of and need for sustainable development.
- 8. Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

On completion of the degree, the students will be able to:

PSO1: Communicate and present civil engineering projects effectively.

PSO2: Use the techniques, Skills and modern engineering tools necessary for civil engineering practice and project management.

PSO3: Provide sustainable solutions to civil engineering problems.

PSO4: Perform as design consultants in construction industry for the design of civil engineering structures.

3.1.1. Course Outcomes (COs) *(SAR should include course outcomes of one course from each semester of study, however, should be prepared for all courses and made available as evidence, if asked) (20)*

Course Code & Title : CE8302 Fluid Mechanics Semester & Year of Study : III Semester & 2020-2021 Course Index : C203	
C203.1	Explain the fundamental concepts of fluid fluids in static, kinematic and dynamic equilibrium.
C203.2	Apply and solve the problems related to the equation of motion.
C203.3	Demonstrate the concepts about dimensional and model analysis.
C203.4	Explain types of flow and losses of flow in pipes.
C203.5	Apply and solve the boundary layer problems

Course Code & Title : CE8402 Strength of Materials II Year / Semester : IV Semester & 2020-2021 Course Index : C212	
C212.1	Determine the strain energy and compute the deflection of determinate beams, frames and trusses using energy principles
C212.2	Analyze propped cantilever, fixed beams and continuous beams using theorem of three moment equation for external loadings and support settlements.
C212.3	Find the load carrying capacity of columns and stresses induced in columns and cylinders.
C212.4	Determine principal stresses and planes for an element in a three dimensional state of stress and study various theories of failure.
C212.5	Determine the stresses due to unsymmetrical bending of beams, locate the shear center, and find the stresses in curved beams.
Course Code & Title : CE8502 Structural Analysis I Semester & Year of study: V Semester & 2021-22 Course Index : C302	
C302.1	Analyze the continuous beams, pin-jointed indeterminate plane frames and rigid plane frames by strain energy method
C302.2	Inspect the continuous beams and rigid frames by slope deflection method.
C302.3	Access the concept of moment distribution and analysis of continuous beams and rigid frames with and without sway.
C302.4	Evaluate the indeterminate pin jointed plane frames, continuous beams and rigid frames using matrix flexibility method.
C302.5	Resolve the concept of matrix stiffness method and analysis of continuous beams, pin jointed trusses and rigid plane frames
Course Code & Title : CE8603 Irrigation Engineering Semester & Year of study: VI Semester & 2021-22 Course Index : C312	
C312.1	Compute the irrigation water requirement of crops using the knowledge and skills acquired through the important terms in Irrigation Engineering.
C312.2	Illustrate the conditions of adopting different methods of irrigation and types of irrigation efficiencies in the field
C312.3	Compare the different types of hydraulic structures for flow diversion and storage with their working principle and its functions
C312.4	Explain about methods of irrigation including canal irrigation
C312.5	Acquire knowledge about various concepts of irrigation water management on optimization of water use

Course Code & Title: CE8702 Railways, Airports, Docks and Harbour Engineering Semester & Year of study: VII Semester & 2022-23 Course Index : C402	
C402.1	Describe the methods of route alignment and design elements in Railway Planning and Constructions.
C402.2	Explain the Construction techniques and Maintenance of Track laying and Railway stations.
C402.3	Review the insights on planning and site selection of Airport Planning and design.
C402.4	Analyze and design the elements for orientation of runways and passenger facility systems.
C402.5	Elaborate the various features in Harbours and Ports, their construction, coastal protection works and coastal Regulations to be adopted.
C402.6	Sketch the layout of railway station and yard available in your residing area.
Course Code & Title : CE8020 Maintenance, Repair and Rehabilitation of Structures Semester & Year of study: VIII Semester & 2022-23 Course Index : C409	
C409.1	Understand the importance of maintenance and assessment method of distressed structures.
C409.2	Enumerate the concept of quality assurance in structures, basic mechanisms by which quality assurance schemes are developed and operated with case studies
C409.3	Acquire the knowledge on recent development in concrete.
C409.4	Suggest the suitable techniques for repair and protection methods
C409.5	Illustrate the repair, rehabilitation and retrofitting of structures and demolition methods.

3.1.2. CO-PO matrices of courses selected in 3.1.1 (six matrices to be mentioned; one per semester from 3rd to 8th semester) **(05)**

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Batch: 2019-2023												
Course Code & Title : CE8302 Fluid Mechanics Semester & Year of Study : III Semester & 2020-2021 Course Index : C203												
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C203.1	3	2		1					1			1
C203.2	3	3	2	2					1			1

C203.3	3	2	1						1			1
C203.4	2	3	2	2	1		2		1			1
C203.5	3	2		2					1			1
C203	3	2.4	1.6	1.75	1		2		1			1
Course Code & Title : CE8402 Strength of Materials II Year / Semester : IV Semester & 2020-2021 Course Index : C212												
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C212.1	3	3	2	1	3				1	2		2
C212.2	3	3	2	1	3				1	2		2
C212.3	3	3	2	1	3				1	2		2
C212.4	3	3	2	1	3				1	2		2
C212.5	3	3	2	1	3				1	2		2
C212	3	3	2	1	3				1	2		2
Course Code & Title : CE8502 Structural Analysis I Semester & Year of study: V Semester & 2021-22 Course Index : C302												
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C302.1	3	3	2	1	3				1	2		2
C302.2	3	3	2	1	3				1	2		2
C302.3	3	3	2	1	3				1	2		2
C302.4	3	3	2	1	3				1	2		2
C302.5	3	3	2	1	3				1	2		2
C302	3	3	2	1	3				1	2		2
Course Code & Title : CE8603 Irrigation Engineering Semester & Year of study: VI Semester & 2021-22 Course Index : C312												
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C312.1	3	2	2			2						1
C312.2	3	2	2		2		1					1
C312.3	3	2	2			2	1	1				1
C312.4	3	2	2			1		1				1
C312.5	2					1			2			1
C312.6	3		1			2		1				
C312	2.83	2	1.8		2	1.6	1	1	2			1
Course Code & Title: CE8702 Railways, Airports, Docks and Harbour Engineering Semester & Year of study: VII Semester & 2022-23 Course Index : C402												
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C402.1	3	2	1	1		1		1		1		1
C402.2	2	2		1		1				1		1
C402.3	2	1	1			1				1		1
C402.4	3	2	3			1		1		1		1
C402.5	2	2	1			1				1		1
C402.6					2					1		1
C402	2.4	1.8	1.5	1	2	1		1		1		1

Course Code & Title : CE8020 Maintenance, Repair and Rehabilitation of Structures Semester & Year of study: VIII Semester & 2022-23 Course Index : C409												
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C409.1	3	2	3				1	1	1			1
C409.2	3		3				1		1			1
C409.3	3	2	3					1	1			1
C409.4	3						1		1			1
C409.5	3		3					1	1			1
C409	3	2	3				1	1	1			1

3.12. B CO-PSO matrices of courses selected in 3.1.1 (six matrices to be mentioned; one per semester from 3rd to 8th semester)

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Batch: 2019-2023				
Course Code & Title : CE8302 Fluid Mechanics Semester & Year of Study : III Semester & 2020-2021 Course Index : C203				
CO	PSO1	PSO2	PSO3	PSO4
C203.1			2	1
C203.2			2	1
C203.3			1	2
C203.4			2	2
C203.5			2	1
C203			1.8	1.4
Course Code & Title : CE8402 Strength of Materials II Year / Semester : IV Semester & 2020-2021 Course Index : C212				
CO	PSO1	PSO2	PSO3	PSO4
C212.1	2	3	2	3
C212.2	2	3	2	3
C212.3	2	3	2	3
C212.4	2	3	2	3
C212.5	2	3	2	3
C212	2	3	2	3
Course Code & Title : CE8502 Structural Analysis I Semester & Year of study: V Semester & 2021-22 Course Index : C302				
CO	PSO1	PSO2	PSO3	PSO4
C302.1	2	3	2	3
C302.2	2	3	2	3
C302.3	2	3	2	3
C302.4	2	3	2	3

C302.5	2	3	2	3
C302.6	2	3	2	3
C302	2	3	2	3
Course Code & Title : CE8603 Irrigation Engineering Semester & Year of study: VI Semester & 2021-22 Course Index : C312				
CO	PSO1	PSO2	PSO3	PSO4
C312.1		2		
C312.2			2	
C312.3	1			2
C312.4	1			1
C312.5		1		
C312.6				2
C312	1	1.5	2	1.67
Course Code & Title: CE8702 Railways, Airports, Docks and Harbour Engineering Semester & Year of study: VII Semester & 2022-23 Course Index : C402				
CO	PSO1	PSO2	PSO3	PSO4
C402.1	2		1	
C402.2		1		
C402.3			2	
C402.4		1		
C402.5				1
C402.6	1			
C402	1.5	1	1.5	1
Course Code & Title : CE8020 Maintenance, Repair and Rehabilitation of Structures Semester & Year of study: VIII Semester & 2023-24 Course Index : C409				
CO	PSO1	PSO2	PSO3	PSO4
C409.1	1	2	1	2
C409.2	1		1	2
C409.3	1	2	1	2
C409.4	1	2	1	2
C409.5	1		1	2
C409	1	2	1	2

3.1.3. (A) Program level Course-PO matrix of all courses INCLUDING first year courses (10)

Course outcomes are statements that specify what a learner will know or be able to do as a result of a learning activity. Outcomes are usually expressed as knowledge, skills, or attitudes. It is a measurable, observable, and specific statement that clearly indicates what a student should know and be able to do as a result of learning. It describes what students are able to demonstrate in terms of knowledge, skills and values upon completion of a course/a span of several courses. Program Outcomes (POs) describe what a program is expected to accomplish. POs describe what students should know and be able to do at the end of the programme. POs are to be in line with the graduate attributes as specified in the NBA.

Program Specific Outcomes (PSOs) are statements that describe what the graduates of a specific engineering program should be able to do. Clear articulation of course outcomes, POs and PSOs serves as the foundation for evaluating the effectiveness of the teaching and learning process.

Course correlation matrix shows the learning relationship (Level of Learning Achieved) between Course Outcomes and Program Outcomes of a course. This matrix strongly indicates whether the students are able to achieve the course objectives/outcomes. The matrix can be used for any course and is a good way to evaluate a course syllabus/content/structure. The below table (Table 3.1 Process for mapping the values for CO-PO Matrix) gives information about the action verbs used in the POs and the nature of POs, stating whether the POs are technical or non-technical, with an understanding of the intention of each PO and the Bloom's level to which each of these action verbs in the POs correlates to. After understanding the POs, write the COs for a course and correlate the COs with the POs.

Table 3.1 Process for mapping the values for CO-PO Matrix

Type	POs	POs Action Verb(s)	POs Bloom's Level(s)	COs Bloom's Level(s)
Technical Skills	PO1	Apply	L3	L1 to L4 »»»» Theory Courses L1 to L5 »»»» Laboratory Courses L1 to L6 »»»» Mini Project and Major Project
	PO2	Identify	L2	
		Formulate	L6	
		Review	L2	
	PO3	Design	L3, L6	
		Develop	L3, L6	
	PO4	Analyze	L4	
		Interpret	L2, L3	
		Design	L6	
	PO5	Create	L6	
		Select	L1, L2, L6	
		Apply	L3	
PO6	Apply	L3		
	Assess	L5		
Transferable Skills	PO7	THUMB RULE		
	PO8	If L1 Action Verbs of a CO »»»»		
	PO9	Correlates with any		
	PO10	of PO7 to PO12 »»»» then assign 1		
	PO11	If L2 to L3 Action Verbs of a CO »»»» Correlates with any		
	PO12	of PO7 to PO12 »»»» then assign 2		
If L4 to L6 Action Verbs of a CO »»»» Correlates with any				
		of PO7 to PO12 »»»» then assign 3		

Note:

- The first six POs are purely technical in nature, while the other POs are transferable skills.
- The CO level is set between 1 and 4 for the theory courses. The CO level is set between 1 and 5 for the laboratory courses. The CO level is extended upto 6th level only for mini projects and major projects.
- For a given course, the course faculty member has to involve all other faculty members who teach that course and ask them to come up with the CO-PO mapping. The course faculty member has to take the average value of all of these CO-PO mappings and finalize the values, or the course faculty member can go with what the majority of the faculty members prefer. While matching COs with non-technical POs, correlate the action verbs used in the COs with the thumb rule given in the table and map the values.

Procedure followed while assigning the values by mapping COs to POs.

- Select action verbs for a CO from different Bloom's levels based on the importance of the particular CO for the given course.
- Stick on to single action verbs while composing COs and use for multiple action verbs if the need arises.
- Values to CO-PO (technical POs in particular) matrix can be assigned by
- Judging the importance of the particular CO in relation to the POs. If the CO matches strongly with a particular PO criterion then 3 is assigned, if it matches moderately then 2 is assigned or less than 1 is assigned else marked with “ - ” symbol.
- If an action verb used in a CO is repeated at multiple Bloom's levels, then reconsider which Bloom's level is the best fit for that action verb.

CO - PO Matrix -BATCH 2019-2023

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) ‘-’: No correlation

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C101									2.2	2.8		
C102	3	2.6	2.2	1					1			1.4
C103	2.4	1.8	1.2	2.4	1.2	2.4					2.2	2.2
C104	2.2	1.2	1								3	3
C105	2.8	1.66	1.75	1	2							1.4
C106	2.8	3			2		2			3		1
C107	3	1.6	2.8		2							1
C108	2.66	3		3				2.66	2.33	2	3	3

C109									2	2.4		
C110	2.2	1.8	2		1							2
C111	2.8	1.8	1	2.8	1	2.8					2.2	2.2
C112	3	2.2	1.8	1.4	-	-	-	-	-	-	-	1
C113			2			2	3	2			2	3
C114	3	2	2.5	1	2	-	-	-	-	-	1	2
C115	1.83	1	1	1			2.83	2	2		2	
C116	3	2	1	3	3	2	1	3	1	2	2	2
C201	3	2	1.2	1.2								1
C202	3	2	-	-	1	-	-	-	-	-	-	-
C203	3	2.4	1.6	1.75	1		2		1			1
C204	1	1.75				1						1
C205			3	3	1	3	1	2				
C206	2.67	1.5	3	1.33	1.5	1.33		1		1	1	1
C207	3	3							3			
C208	3			2	3				2			
C209									2.6	3		
C210	1.8	1.8	1.8	1.2	1						1	
C211	2.17	2.2	2.33	1.67	1.8	2	3	2		1		1
C212	3	3	2	1	3				1	2		2
C213	3	2.8	2.2	1.4	1							1
C214	3	2	2	1.5		1.3	1.2					1
C215	3	2	1	1		2	1	0	0	0	1	1
C216	3	2	1	3	3	2	1	1	1	1	2	2
C217	3	2	2	2								
C218									2	2.8		
C301	1.5	2.1	2.5	1	2.1							
C302	3	3	2	1	3				1	2		2
C303	3	2	3	2		2.3	2					2
C304	2	2.5	2	3						1		
C305	2	2	2	-	-	3	-	-	1	2	-	1
C306	2	1			2		2	1	2.3	2		1

C307	3	2	2	2	2	2		2	1			2
C308	3			2	3		1.5		2	2		
C309	3	2	2	2	3	3		2	3	2		2
C310	2	2	3	2	2	2	2	2	2	1	2	2
C311	3	3	2	1	3				1	2		2
C312	2.83	2	1.8		2	1.6	1	1	2			1
C313	2.5	3	2.8	1.5		2.5	1	2.25	2		2	
C314	3	2	3	2		1	2		1			2
C315	2.5	1.5	2	-	-	1.5	2.16	-	-	-	-	-
C316	3	-	-	1	-	-	-	-	3	-	-	-
C317	2	2	2				1			1		
C318									2.5	2.6		
C401	2	1.7			1.5	0.3					1.6	1.5
C402	2.4	1.8	1.5	1	2	1		1		1		1
C403	2	2	3	2	2	2	2	2	2	1	2	2
C404	2	1	2			2	2		2	1.25		1.5
C405	2	2	2	2			2					1
C406	3	3	3	3	3	2	2	1	3	3	2	2
C407	3	3	3	3	3	2	2	1	3	3	2	2
C408	1.75	1.5	1			3		3	3	3		1
C409	3	2	3				1	1	1			1
C410	3	3	3	3	2	2	2	1	3	3	2	2
PO Average	2.6	2.1	2.1	1.8	2.0	2.0	1.8	1.6	1.9	1.9	1.9	1.6

3.1.3. (B) Program level Course-PSO matrix of all courses INCLUDING first year courses (10)

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) '-' : No correlation

Course	PSO1	PSO2	PSO3	PSO4
C101				
C102		1	1.5	
C103	1.2		2.4	1.2

C104	1			
C105		1	1.4	
C106	2			
C107	1	1	1	1
C108	1.66		1	1
C109				
C110		1.33	1	
C111	1		2.8	1
C112	2.2	2	1	-
C113			1	
C114	-	1	-	-
C115				
C116	2	3	2	3
C201			1.5	2
C202	-	1	-	-
C203			1.8	1.4
C204		1	1	
C205		3	2	
C206	1.17	1.33	2	1.67
C207	2.8	3		3
C208		2	2	3
C209			1.5	2
C210			1.25	
C211	1.5	2	1	1.33
C212	2	3	2	3
C213	1.5		1.6	2
C214	1	2	2	1
C215	1	1	1.2	1.33
C216	2	3	2	3
C217			3	
C218				
C301		1.6		2.83

C302	2	3	2	3
C303	1	3	2.75	2.5
C304	1	1	2	2
C305	-	1	1	-
C306			1	
C307	2	3	2	3
C308	2	2	2	
C309	2	2	2	2
C310	2	3	2	3
C311	2	3	2	3
C312	1	1.5	2	1.67
C313	2	2	1.8	1.5
C314	1	3	2.75	2.5
C315	-	1	-	1
C316	2.8	-	-	3
C317	1		1	2
C318				
C401		1.5		
C402	1.5	1	1.5	1
C403	2	3	2	3
C404	1.5	1.5	1.83	-
C405		1.66		
C406	3	2	2	2
C407	3	2	2	2
C408				2
C409	1	2	1	2
C410	3	2	2	2
PSO Average	1.7	1.9	1.7	2.1

3.2. Attainment of Course Outcomes (50)

3.2.1. Describe the assessment processes used to gather the data upon which the evaluation of Course Outcome is based (10)

3.2.1 (A) Assessment processes of CO attainment

For measuring the attainment of Course Outcomes, various tools are used. The process of CO attainment is described in Figure 3.1.

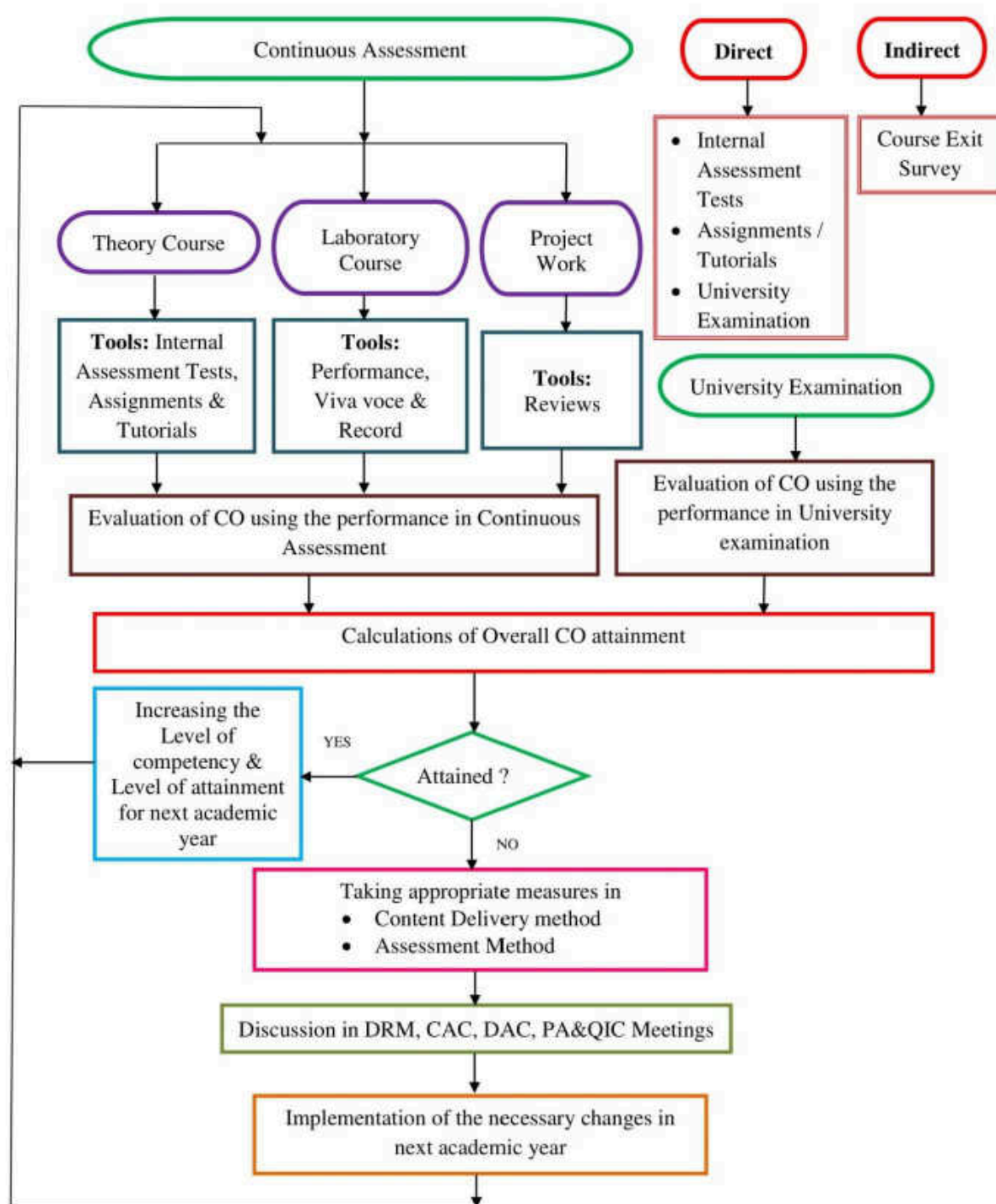
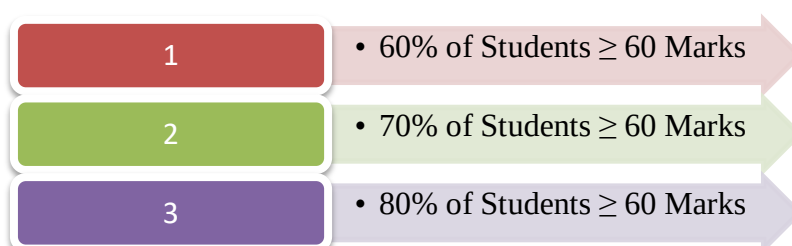


Figure 3.1 Assessment process of CO attainment

Direct Assessment:

<i>Evaluation Methods</i>	<i>Process</i>
Internal Assessment Tests	To evaluate the attainment of course outcomes, three internal assessment tests are conducted per semester for R2017 and two per semester for R2021. Each question is mapped with COs and blooms levels.
Assignments & Tutorials	The tutorials and assignments are given to the students based on the subject nature. Tutorial and Assignment sheets are prepared by the faculty member with COs and levels.
Continuous Assessment & Model Exam (Laboratory Course)	The evaluation criteria for each experiment are based on performance, viva-voce, and record mark. The attainment of COs is calculated through continuous assessment and model practical performance.
Project Reviews	• Three reviews are conducted periodically to monitor and evaluate the progress of the project using project rubrics. • Viva-Voce is conducted at the end of the semester as per university norms.
University Examination	At the end of each semester, final examination is conducted for Theory and Laboratory courses by Anna University, in which question paper covers the entire syllabus and all the COs are covered in the question papers.

Attainment Level:**Theory courses:**

For each theory course, faculty member calculates the course outcome attainment using the University Examination and Internal Assessment Test. The attainment level will be calculated based on the average performance levels of both the University Examination and Internal Assessment Test. The evaluation process of Internal Assessment Tests/Assignments/Tutorials/Quiz is counted for 40% and the remaining 60% will be given

for university examination. Based on the level of CO attainment, the faculty member will decide whether to increase the competency level or change the content delivery method, assessment methods to improve attainment level for the course.

Assessment Tool		Weightage	Frequency
CO Attainment	Internal Assessment Tests	40%	Thrice in a Semester for R2017 and Twice in a Semester for R2021
	University Examination	60%	Once in a Semester

Laboratory Courses:

For laboratory courses, the course outcome will be calculated based on performance, viva-voce, record work and model practical examination with the weightage of 40% for Continuous Internal Assessment and 60 % weightage for University Practical Examination. Based on the CO attainment level, the faculty member will decide whether to increase the competency level or enhance the practical knowledge of the students in order to improve attainment level for the laboratory course.

Assessment Tool		Weightage	Frequency
CO Attainment	Continuous Internal Assessment	40%	Every Week
	University Practical Examination	60%	Once in a Semester

Project Work Assessment:

For project work, Continuous Internal Assessment is based on the performance in the three reviews. The Course Attainment is calculated based on the three reviews and project Viva voce.

- Project review is conducted every month to review the progress of the students and the second review will be conducted in the presence of an industry expert.
- Suggestions are given to the students for their continuous update and improvement.
- Evaluation of each review are based on the parameters discussed in teaching learning process.

The faculty member will decide the competency level and attainment level for project work considering the average performance level of the students.

Assessment Tool		Weightage	Frequency
Continuous Assessment	Reviews	40%	Every Month
University Assessment	Viva-Voce	60%	Once in a Semester

CO Attainment Calculation:

The course outcomes for all the courses are calculated in terms of percentage using the formula.

$$\text{COx in \%} = \frac{\text{Marks obtained by the students in COx}}{\text{Maximum Marks allotted in COx}} \times 100$$

Where, x= [1 to N], N= Number of COs.

Each course outcome is calculated for all the students based on marks obtained by the students.

$$\text{COx Attainment in \%} = \frac{\text{no. of Students scored more than or equal to 60\% of Marks in COx}}{\text{no. of Students}} \times 100$$

Where, x= [1 to N], N= Number of Course Outcomes

CO Attainment level is defined based on the following criteria:

COx Attainment Level	3	80% of the Students scoring more than or equal to 60% of Marks in COx
	2	70% of the Students scoring more than or equal to 60% of Marks in COx
	1	60% of the Students scoring more than or equal to 60% of Marks in COx

After calculating the attainment levels of each COs from the performance of Internal Assessment Test 1, 2 & 3, the attainment level of internal assessment test is calculated with ratio of sum of all COs attained by total number of COs as shown below:

$$\text{IAT Attainment Level} = \frac{\text{Sum of all COs attained by students}}{\text{Total Number of COs}}$$

Based on university grade, the attainment level of COs is calculated. The attainment level is decided based on the following criteria.

University Attainment Level	3	80% of the students scoring more than or equal to 60% of Marks in University Exam
	2	70% of the students scoring more than or equal to 60% of Marks in University Exam

	1	60% of the students scoring more than or equal to 60% of Marks in University Exam
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Overall CO Attainment:

The Overall Attainment for a course is sum of 40% of Internal Assessment Test Attainment Level and 60% of the University Attainment Level.

$$\text{Overall CO Attainment} = \left(\frac{\sum_{i=1}^n CO_i}{n} \times 0.4 \right) + (UA \times 0.6)$$

Where n= number of course outcomes.

The weightage of the assessment varies based on the nature of the course and assessment methods used for that particular course by course-handling faculty members.

Sample Course for Calculation of CO Attainment:

Course Code & Title : EN8592 - Wastewater Engineering
Semester & Year of study : VI Semester, III Year – Batch 2020-2024
Index : C315

Name of the Subject: EN8592 - Wastewater Engineering
Name of the Faculty: SUBHA C
Year/Semester: III / VI - CIVIL

Detailed Report - CIVIL																																			
CO1						CO2						CO3						CO4						CO5						CO6					
Sl.No	Name of the student / Roll No.	IAT 60%	Assignment 10%	Tutorial 20%	CO End Survey 10%	CO1 Total	IAT 60%	Assignment 10%	Tutorial 20%	CO End Survey 10%	CO2 Total	IAT 60%	Assignment 10%	Tutorial 20%	CO End Survey 10%	CO3 Total	IAT 60%	Assignment 10%	Tutorial 20%	CO End Survey 10%	CO4 Total	IAT 60%	Assignment 10%	Tutorial 20%	CO End Survey 10%	CO5 Total	Assignment 50%	CO End Survey 10%	CO6 Total	Univ					
1	953620103001	23/33	15/20	15/20	100.0	74.32	15/34	15/20	16/20	100.0	59.97	2/33	15/20	15/20	100.0	37.64	38/55	17/20	17/20	100.0	74.51	5/42	15/20	15/20	100.0	43.93	16/20	100.0	82	0					
2	953620103002	16/33	16/20	15/20	100.0	62.09	19/34	16/20	16/20	100.0	65.76	17/33	17/20	15/20	100.0	64.41	20/55	18/20	16/20	100.0	55.89	5/42	16/20	15/20	100.0	40.14	16/20	100.0	82	70					
3	953620103003	24/33	16/20	15/20	100.0	76.64	27/34	16/20	16/20	100.0	81.65	26/33	17/20	16/20	100.0	81.77	48/57	17/20	17/20	100.0	86.03	18/43	17/20	15/20	100.0	58.62	17/20	100.0	86.5	60					
4	953620103004	17/33	15/20	16/20	100.0	64.41	26/34	17/20	16/20	100.0	80.38	17/33	17/20	17/20	100.0	66.41	20/55	16/20	17/20	100.0	55.89	4/42	17/20	14/20	100.0	38.21	17/20	100.0	86.5	60					
5	953620103005	10/33	14/20	17/20	100.0	82.18	23/34	17/20	17/20	100.0	76.09	2/33	17/20	17/20	100.0	39.14	3/55	15/20	17/20	100.0	37.6	1/42	15/20	15/20	100.0	33.93	16/20	100.0	82	60					
6	953620103006	17/33	15/20	17/20	100.0	85.41	27/34	16/20	16/20	100.0	81.65	16/33	18/20	16/20	100.0	84.09	41/55	16/20	17/20	100.0	77.41	15/42	15/20	16/20	100.0	54.93	15/20	100.0	77.5	60					
7	953620103007	4/33	15/20	17/20	100.0	41.77	16/34	16/20	17/20	100.0	63.24	4/33	15/20	17/20	100.0	43.27	23/55	16/20	16/20	100.0	57.79	1/42	16/20	16/20	100.0	35.43	15/20	100.0	77.5	60					
8	953620103008	31/33	19/20	18/20	100.0	93.86	33/34	19/20	20/20	100.0	97.24	31/33	19/20	19/20	100.0	93.86	50/55	19/20	19/20	100.0	89.72	36/42	19/20	20/20	100.0	90.43	20/20	100.0	100	70					
9	953620103009	31/33	19/20	18/20	100.0	93.86	33/34	19/20	20/20	100.0	97.24	30/33	19/20	19/20	100.0	92.05	53/55	18/20	19/20	100.0	92.83	40/42	19/20	20/20	100.0	96.64	20/20	100.0	100	80					
10	953620103010	28/33	17/20	17/20	100.0	86.41	18/34	17/20	16/20	100.0	82.74	3/33	18/20	16/20	100.0	40.45	8/55	15/20	16/20	100.0	41.78	0/42	16/20	17/20	100.0	35	16/20	100.0	82	70					
11	953620103301	3/33	15/20	16/20	100.0	38.95	9/34	17/20	16/20	100.0	50.38	0/33	17/20	16/20	100.0	34.5	1/55	14/20	14/20	100.0	32.03	0/42	15/20	16/20	100.0	33.5	17/20	100.0	86.5	0					
12	953620103302	18/33	17/20	16/20	100.0	67.23	19/34	16/20	16/20	100.0	65.76	7/33	15/20	16/20	100.0	46.23	7/55	14/20	14/20	100.0	38.24	0/42	15/20	15/20	100.0	32.5	16/20	100.0	82	60					
13	953620103303	8/33	16/20	16/20	100.0	48.55	9/34	17/20	16/20	100.0	48.62	2/33	16/20	17/20	100.0	38.64	6/55	13/20	15/20	100.0	37.71	3/42	15/20	14/20	100.0	35.79	16/20	100.0	82	60					
14	953620103304	21/33	18/20	16/20	100.0	73.18	30/34	18/20	17/20	100.0	88.94	29/33	18/20	17/20	100.0	88.73	32/57	18/20	18/20	100.0	70.68	36/43	18/20	17/20	100.0	86.23	15/20	100.0	91	70					
15	953620103305	15/33	16/20	16/20	100.0	61.27	14/34	17/20	17/20	100.0	60.21	4/33	17/20	16/20	100.0	41.77	10/55	16/20	16/20	100.0	44.34	0/42	17/20	18/20	100.0	36.5	17/20	100.0	86.5	0					
16	953620103307	18/33	18/20	16/20	100.0	67.73	29/34	17/20	19/20	100.0	87.68	19/33	17/20	16/20	100.0	67.23	44/55	16/20	18/20	100.0	81.52	16/42	16/20	19/20	100.0	59.86	17/20	100.0	86.5	60					
17	953620103308	22/33	18/20	16/20	100.0	75	24/34	18/20	19/20	100.0	79.35	15/33	18/20	18/20	100.0	64.27	24/55	17/20	18/20	100.0	61.33	19/42	16/20	16/20	100.0	61.14	18/20	100.0	91	60					
18	953620103309	16/33	15/20	16/20	100.0	62.59	17/34	16/20	17/20	100.0	65	3/33	15/20	16/20	100.0	38.95	5/55	17/20	15/20	100.0	38.67	0/42	15/20	17/20	100.0	34.5	17/20	100.0	86.5	60					
19	953620103310	19/33	15/20	17/20	100.0	69.05	26/34	15/20	16/20	100.0	79.38	16/33	15/20	17/20	100.0	63.59	30/57	15/20	16/20	100.0	65.08	4/43	15/20	17/20	100.0	40.08	17/20	100.0	86.5	60					
20	953620103311	13/33	15/20	16/20	100.0	57.14	20/34	15/20	16/20	100.0	68.79	6/33	16/20	16/20	100.0	44.91	28/55	14/20	17/20	100.0	62.97	0/42	16/20	16/20	100.0	34	17/20	100.0	86.5	0					
21	953620103312	16/33	17/20	16/20	100.0	63.59	25/34	15/20	16/20	100.0	77.62	14/33	16/20	16/20	100.0	59.45	32/55	15/20	17/20	100.0	67.6	0/42	16/20	16/20	100.0	34	17/20	100.0	86.5	60					
Number of Students Scored > 60 in CIA&I and > 60 in Univ & Percentage			16 / 76.19%					18 / 85.71%					10 / 47.62%					11 / 52.38%					4 / 19.05%					21 / 100%					17 / 80.95%		
Attainment			2					3					0					0					0					3					3		
Overall Course Attainment : 2.33																																			

Figure 3.2 Sample – Course Attainment – EN8592 Waste Water Engineering VI Semester

The course outcomes attainment for the sample course taken above is calculated as per the table mentioned below.

Continuous Internal Assessment Methods (CIAM)	Weightage							
	CO1	CO2	CO3	CO4	CO5	CO6	CIAM	End Semester
IAT	60%	60%	60%	60%	60%		40%	60%
Assignment	10%	10%	10%	10%	10%	90%		
Tutorial	20%	20%	20%	20%	20%			
CO End Survey	10%	10%	10%	10%	10%	10%		
Total	100 %	100 %	100 %	100 %	100 %	100 %	100%	

Sample Calculation:

$$\text{COx in \%} = \frac{\text{Marks obtained by the students in COx}}{\text{Maximum Marks allotted in COx}} \times 100$$

Where, x= [1 to N], N= Number of COs

Continuous Internal Assessment Methods (CIAM)	Reg. No of the student: 953620103001					
	CO1	CO2	CO3	CO4	CO5	CO6
IAT (60% Weightage)	CO1 in % = $\frac{23}{33}$ =69.67	CO2 in % = $\frac{15}{34}$ =44.12	CO3 in % = $\frac{2}{33}$ = 6.06	CO4 in % = $\frac{38}{58}$ = 65.52	CO5 in % = $\frac{8}{42}$ = 19.05	-
Assignment (10% Weightage)	CO1 in % = $\frac{15}{20}$ =75	CO2 in % = $\frac{15}{20}$ =75	CO3 in % = $\frac{18}{20}$ = 90	CO4 in % = $\frac{17}{20}$ = 85	CO5 in % = $\frac{15}{20}$ = 75	CO6 in % = $\frac{16}{20}$ = 80
Tutorial (20% Weightage)	CO1 in % = $\frac{15}{20}$ =75	CO2 in % = $\frac{16}{20}$ =80	CO3 in % = $\frac{15}{20}$ = 75	CO4 in % = $\frac{17}{20}$ = 85	CO5 in % = $\frac{15}{20}$ = 75	-
CO End Survey (10% Weightage)	100	100	100	100	100	100
Total	74.32	59.97	37.64	74.81	43.93	82
Continuous Internal Assessment Methods (CIAM)	Reg. No of the student: 953620103008					
	CO1	CO2	CO3	CO4	CO5	CO6
IAT (60% Weightage)	CO1 in % = $\frac{31}{33}$ =93.93	CO2 in % = $\frac{33}{34}$ = 97.05	CO3 in % = $\frac{31}{33}$ = 93.93	CO4 in % = $\frac{50}{58}$ = 86.20	CO5 in % = $\frac{36}{42}$ = 85.71	-
Assignment (10% Weightage)	CO1 in % = $\frac{19}{20}$ = 95	CO2 in % = $\frac{18}{20}$ = 90	CO3 in % = $\frac{19}{20}$ = 95	CO4 in % = $\frac{18}{20}$ = 90	CO5 in % = $\frac{18}{20}$ = 90	CO6 in % = $\frac{20}{20}$ = 100
Tutorial (20% Weightage)	CO1 in % = $\frac{18}{20}$ =90	CO2 in % = $\frac{20}{20}$ = 100	CO3 in % = $\frac{18}{20}$ = 90	CO4 in % = $\frac{19}{20}$ = 95	CO5 in % = $\frac{20}{20}$ = 100	-
CO End Survey (10% Weightage)	100	100	100	100	100	100
Total	93.86	97.23	93.86	89.72	90.43	100

Continuous Internal Assessment Methods (CIAM)	Reg. No of the student: 953620103303					
	CO1	CO2	CO3	CO4	CO5	CO6
IAT (60% Weightage)	$\text{CO1 in \%} = \frac{8}{33}$ = 24.24	$\text{CO2 in \%} = \frac{8}{34}$ = 23.52	$\text{CO3 in \%} = \frac{2}{33}$ = 6.06	$\text{CO4 in \%} = \frac{6}{58}$ = 10.34	$\text{CO5 in \%} = \frac{3}{42}$ = 7.14	-
Assignment (10% Weightage)	$\text{CO1 in \%} = \frac{16}{20}$ = 80	$\text{CO2 in \%} = \frac{17}{20}$ = 85	$\text{CO3 in \%} = \frac{16}{20}$ = 80	$\text{CO4 in \%} = \frac{13}{20}$ = 65	$\text{CO5 in \%} = \frac{15}{20}$ = 75	$\text{CO6 in \%} = \frac{16}{20}$ = 80
Tutorial (20% Weightage)	$\text{CO1 in \%} = \frac{16}{20}$ = 80	$\text{CO2 in \%} = \frac{16}{20}$ = 80	$\text{CO3 in \%} = \frac{17}{20}$ = 85	$\text{CO4 in \%} = \frac{15}{20}$ = 75	$\text{CO5 in \%} = \frac{14}{20}$ = 70	-
CO End Survey (10% Weightage)	100	100	100	100	100	100
Total	48.55	48.62	38.64	37.71	35.79	82

We used the above formula to calculate CO attainment in terms of percentage for all the students. The CO attainment level is calculated by as follows

CO1	2	76% of Students scored more than 60 Marks
CO2	3	86% of Students scored more than 60 Marks
CO3	0	48% of Students scored more than 60 Marks
CO4	0	52% of Students scored more than 60 Marks
CO5	0	19% of Students scored more than 60 Marks
CO6	3	100% of Students scored more than 60 Marks

Internal Attainment is calculated as follows:

$$\text{IAT Attainment Level} = \frac{\text{CO1} + \text{CO2} + \text{CO3} + \text{CO4} + \text{CO5} + \text{CO6}}{5}$$

$$\text{IAT Attainment Level} = \frac{2 + 3 + 0 + 0 + 0 + 3}{6} = 1.33$$

University Attainment is calculated as follows:

The university attainment level can be calculated as follows:

University Attainment level	1	60% of Students scored more than or equal to 60 Marks
	2	70% of Students scored more than or equal to 60 Marks
	3	80% of Students scored more than or equal to 60 Marks

In this subject 80.95% of the students scored 60% of the marks in university examination, so the University Attainment level is 3

Overall Attainment is calculated as follows:

The Overall Attainment for the course is calculated as follows.

$$\text{Overall CO Attainment} = \left(\frac{\sum_{i=1}^n CO_i}{n} \times 0.4 \right) + (UA \times 0.6)$$

Where n = number of course outcome.

$$\text{Overall Attainment Level} = (0.4 \times 1.33) + (0.6 \times 3) = 2.33$$

$$\text{Overall Attainment Level} = 2.33$$

3.2.2 Record the attainment of Course Outcomes of all courses with respect to set attainment levels (40)

The attainment level for each course is decided by the respective faculty member. The attainment of COs for all subjects from the I year, II-year, III year, and IV year for the batch 2018-2022, 2019-2023, and 2020-2024 are assessed by having 60% weightage for university examination and 40% weightage to Continuous Internal Assessment Methods (CIAM) which includes Internal Assessment Tests and Assignments/Tutorial/Quiz/Mini Project and Course Outcome End Survey.

Set Attainment Level Calculation:

- The set attainment level for the first batch (2017-2021) has been fixed as 1.8 for theory courses and 2.1 for Laboratory courses/Project.
- The set attainment level for the batch 2018-2022 has been fixed by taking the average grade point analysis value of the university examinations obtained by the 2017-2021 batch students.
- The set attainment level for the batch 2019-2023 has been fixed by taking the average value of average grade point analysis value of the university examination obtained by the previous two batch students (2017-2021 & 2018-2022)

The table 3.2 shows the methodology of Target attainments (Set Attainment Level) for all the courses.

Table 3.2 Targets for Course outcomes

Batch	Target Attainment
2017-2021 (AGPA ₁)	1.8 for Theory Courses & 2.1 for Laboratory Courses/Projects
2018-2012 (AGPA ₂)	AGPA ₁ × 0.3
2019-2023 (AGPA ₃)	$\left(\frac{(AGPA_1 + AGPA_2)}{2} \right) \times 0.3$

2020-2024	$\left(\frac{AGPA_1 + AGPA_2 + AGPA_3}{3} \right) \times 0.3$
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BATCH: 2018-2022

2018-2022 Batch					
S.No	Course Index	Course Code & Course Name	Target	CO Attainment	Attainment Status
2018-2019 Odd Semester I Semester					
1.	C101	HS8151 - Communicative English	2.20	2.91	Attained
2.	C102	MA8151 - Engineering Mathematics I	1.80	1.31	Not Attained
3.	C103	PH8151 - Engineering Physics	1.80	2.71	Attained
4.	C104	CY8151 - Engineering Chemistry	2.04	3.00	Attained
5.	C105	GE8151 - Problem Solving and Python Programming	1.92	2.03	Attained
6.	C106	GE8152 - Engineering Graphics	2.34	2.57	Attained
7.	C107	GE8161 - Problem Solving and Python Programming Laboratory	2.70	2.68	Not Attained
8.	C108	BS8161 - Physics and Chemistry Laboratory	2.70	3.00	Attained
2018-2019 Even Semester II Semester					
9.	C109	HS8251 - Technical English	2.16	3.00	Attained
10.	C110	MA8251 - Engineering Mathematics II	2.16	2.57	Attained
11.	C111	PH8201 - Physics of Civil Engineering	2.52	2.78	Attained
12.	C112	BE8251 - Basic Electrical and Electronics Engineering	1.80	2.06	Attained
13.	C113	GE8291 - Environmental Science and Engineering	2.13	2.71	Attained
14.	C114	GE8292 - Engineering Mechanics	1.80	2.82	Attained
15.	C115	GE8261 - Engineering Practices Laboratory	2.79	3.00	Attained
16.	C116	CE8211 - Computer Aided Building Drawing	2.76	3.00	Attained
2019-2020 Odd Semester III Semester					
17.	C201	MA8353 - Transforms and Partial Differential Equations	1.8	1.67	Not Attained
18.	C202	CE8301 - Strength of Material I	1.8	2.24	Attained
19.	C203	CE8302 - Fluid Mechanics	1.8	3.00	Attained
20.	C204	CE8351 - Surveying	1.8	2.32	Attained
21.	C205	CE8391 - Construction Materials	1.8	2.46	Attained
22.	C206	CE8392 - Engineering Geology	1.92	0.30	Attained
23.	C207	CE8311 - Construction Materials Laboratory	2.97	3.00	Attained
24.	C208	CE8361 - Surveying Laboratory	2.61	3.00	Attained
25.	C209	HS8381 - Interpersonal Skills / Listening and Speaking	2.37	3.00	Attained

2019-2020 Even Semester IV Semester					
26.	C210	MA8491 - Numerical Methods	1.80	2.73	Attained
27.	C211	CE8401 - Construction Techniques and Practices	2.04	3.00	Attained
28.	C212	CE8402 - Strength of Materials II	1.80	2.60	Attained
29.	C213	CE8403 - Applied Hydraulic Engineering	1.80	2.92	Attained
30.	C214	CE8404 - Concrete Technology	1.80	2.76	Attained
31.	C215	CE8491 - Soil Mechanics	1.80	2.35	Attained
32.	C216	CE8481 - Strength of Materials Laboratory	2.64	3.00	Attained
33.	C217	CE8461 - Hydraulic Engineering Laboratory	2.73	3.00	Attained
34.	C218	HS8461 - Advanced Reading and Writing	2.52	3.00	Attained
2020-2021 Odd Semester V Semester					
35.	C301	CE8501 - Design of Reinforced Cement Concrete Elements	1.80	3.00	Attained
36.	C302	CE8502 - Structural Analysis I	2.07	3.00	Attained
37.	C303	EN8491 - Water Supply Engineering	2.25	3.00	Attained
38.	C304	CE8591 - Foundation Engineering	1.80	3.00	Attained
39.	C305	ORO551 - Renewable Energy Sources	2.25	3.00	Attained
40.	C306	OCH551 - Industrial Nanotechnology	2.07	2.92	Attained
41.	C307	GE8071 - Disaster Management	1.8	3.00	Attained
42.	C308	CE8511 - Soil Mechanics Laboratory	2.88	3.00	Attained
43.	C309	CE8512 - Water and Waste Water Laboratory	2.88	3.00	Attained
44.	C310	CE8513 - Survey Camp	2.61	3.00	Attained
2020-2021 Even Semester VI Semester					
45.	C311	CE8601 - Design of Steel Structural Elements	2.49	3.00	Attained
46.	C312	CE8602 - Structural Analysis II	2.43	3.00	Attained
47.	C313	CE8603 - Irrigation Engineering	2.43	2.84	Attained
48.	C314	CE8604 - Highway Engineering	2.25	2.84	Attained
49.	C315	EN8592 - Wastewater Engineering	2.25	3.00	Attained
50.	C316	CE8005 - Air Pollution and Control Engineering	2.76	3.00	Attained
51.	C317	CE8611 - Highway Engineering Laboratory	2.97	3.00	Attained
52.	C318	CE8612 - Irrigation and Environmental Engineering Drawing	2.88	3.00	Attained
53.	C319	HS8581 - Professional Communication	2.85	3.00	Attained
2021-2022 Odd Semester VII Semester					

54.	C401	CE8701 - Estimation, Costing and valuation Engineering	2.40	2.87	Attained
55.	C402	CE8702 - Railways, Airports, Docks and Harbour Engineering	2.28	2.67	Attained
56.	C403	CE8703 - Structural Design and Drawing	1.95	2.64	Attained
57.	C404	CE8012 - Construction Planning and Scheduling	1.80	2.73	Attained
58.	C405	EN8591 - Municipal Solid Waste Management	2.37	3.00	Attained
59.	C406	OML751 - Testing of Materials	2.52	3.00	Attained
60.	C407	OCS752 - Introduction to C Programming	2.40	2.07	Attained
61.	C408	CE8711 - Creative Innovative Project	3.00	3.00	Attained
62.	C409	CE8712 - Industrial Training	2.97	3.00	Attained
2021-2022 Even Semester VIII Semester					
63.	C410	GE8076 - Professional Ethics in Engineering	2.43	2.68	Attained
64.	C411	CE8020 - Maintenance, Repair and Rehabilitation of Structures	2.58	2.80	Attained
65.	C412	CE8811 - Project Work	3.00	3.00	Attained

BATCH: 2019-2023

2019-2023 Batch					
S.No	Course Index	Course Code & Course Name	Target	CO Attainment	Attainment Status
2019-2020 Odd Semester I Semester					
1.	C101	HS8151 - Communicative English	2.2	3	Attained
2.	C102	MA8151 - Engineering Mathematics I	1.8	2.64	Attained
3.	C103	PH8151 - Engineering Physics	1.83	2.21	Attained
4.	C104	CY8551 - Engineering Chemistry	2.06	2.64	Attained
5.	C105	GE8151 - Problem Solving and Python Programming	1.83	1.2	Not Attained
6.	C106	GE8152 - Engineering Graphics	2.34	2.64	Attained
7.	C107	GE8161 - Problem Solving and Python Programming Laboratory	2.7	3	Attained
8.	C108	BS8161 - Physics and Chemistry Laboratory	2.7	3	Attained
2019-2020 Even Semester II Semester					
9.	C109	HS8251 - Technical English	2.16	2.8	Attained
10.	C110	MA8251 - Engineering Mathematics – II	2.16	2.2	Attained

11.	C111	PH8201 - Physics for Civil Engineering	2.76	2.33	Not Attained
12.	C112	BE8251 - Basic Electrical and Electronics Engineering	1.8	2.6	Attained
13.	C113	GE8291 - Environmental Science and Engineering	2.13	3	Attained
14.	C114	GE8292 - Engineering Mechanics	1.8	2.84	Attained
15.	C115	GE8261 - Engineering Practices Laboratory	2.79	3	Attained
16.	C116	CE8211 - Computer Aided Building Drawing	2.76	3	Attained
2020-2021 Odd Semester III Semester					
17.	C201	MA8353 - Transforms and Partial Differential Equations	1.8	2.68	Attained
18.	C202	CE8301 - Strength of Materials – I	1.8	2.76	Attained
19.	C203	CE8302 - Fluid Mechanics	1.8	3	Attained
20.	C204	CE8351 - Surveying	1.8	3	Attained
21.	C205	CE8391 - Construction Materials	1.8	2.84	Attained
22.	C206	CE8392 - Engineering Geology	1.92	2.84	Attained
23.	C207	CE8311 - Construction Materials Laboratory	2.97	3	Attained
24.	C208	CE8361 - Surveying Laboratory	2.66	3	Attained
25.	C209	HS8381 - Interpersonal Skills / Listening and Speaking	2.37	3	Attained
2020-2021 Even Semester IV Semester					
26.	C210	MA8491 - Numerical Methods	1.8	2.92	Attained
27.	C211	CE8401 - Construction Techniques and Practices	2.15	3	Attained
28.	C212	CE8402 - Strength of Materials II	1.8	3	Attained
29.	C213	CE8403 - Applied Hydraulic Engineering	1.93	3	Attained
30.	C214	CE8404 - Concrete Technology	1.9	3	Attained
31.	C215	CE8491 - Soil Mechanics	1.8	3	Attained
32.	C216	CE8481 - Strength of Materials Laboratory	2.73	3	Attained
33.	C217	CE8461 - Hydraulic Engineering Laboratory	2.8	3	Attained
34.	C218	HS8461 - Advanced Reading and Writing	2.63	3	Attained
2021-2022 Odd Semester V Semester					
35.	C301	CE8501 - Design of Reinforced Cement concrete Elements	1.82	2.67	Attained
36.	C302	CE8502 - Structural Analysis I	2.13	2.58	Attained

37.	C303	EN8491 - Water and Waste Water Engineering	2.25	3	Attained
38.	C304	CE8591 - Foundation Engineering	1.95	2.88	Attained
39.	C305	GE8071 - Disaster Management	2.16	2.47	Attained
40.	C306	ORO551 - Renewable Energy Sources	2.23	2.8	Attained
41.	C307	CE8511 - Soil Mechanics Laboratory	2.88	3	Attained
42.	C308	CE8512 - Water and Waste Water Analysis Laboratory	2.88	3	Attained
43.	C309	CE8513 - Survey Camp	2.69	3	Attained
2021-2022 Even Semester VI Semester					
44.	C310	CE8601 - Design of Steel Structures	2.53	2.28	Not Attained
45.	C311	CE8602 - Structural Analysis II	2.64	2.76	Attained
46.	C312	CE8603 - Irrigation Engineering	2.56	2.2	Not Attained
47.	C313	CE8604 - Highway Engineering	2.32	2.12	Not Attained
48.	C314	EN8592 - Wastewater Engineering	2.39	2.73	Attained
49.	C315	CE8005 - Air Pollution and Control Engineering (Elective II)	2.65	2.8	Attained
50.	C316	CE8611 - Highway Engineering Laboratory	2.97	3	Attained
51.	C317	CE8612 - Irrigation and Environmental Engineering Drawing	2.89	3	Attained
52.	C318	HS8581 - Professional Communication	2.85	3	Attained
2022-2023 Odd Semester VII Semester					
53.	C401	CE8701 - Estimation, Costing and valuation Engineering	2.4	2.73	Attained
54.	C402	CE8702 - Railways, Airports, Docks and Harbour Engineering	2.35	2.93	Attained
55.	C403	CE8703 - Structural Design and Drawing	2.09	2.64	Attained
56.	C404	EN8591 - Municipal Solid Waste Management	2.42	2.93	Attained
57.	C405	OML751 - Testing of Materials	2.58	2.92	Attained
58.	C406	CE8711 - Creative Innovative Project	3	3	Attained
59.	C407	CE8712 - Industrial Training	2.98	3	Attained
2022-2023 Even Semester VIII Semester					

60.	C408	GE8076 - Professional Ethics in Engineering	2.43	2.52	Attained
61.	C409	CE8020 - Maintenance, Repair and Rehabilitation of Structures	2.58	2.92	Attained
62.	C410	CE8811 - Project Work	3	3	Attained

BATCH: 2020-2024

2020-2024 Batch					
S.No	Course Index	Course Code & Course Name	Target	CO Attainment	Attainment Status
2020-2021 Odd Semester I Semester					
1.	C101	HS8151 - Communicative English	2.2	3	Attained
2.	C102	MA8151 - Engineering Mathematics-I	1.8	2.57	Attained
3.	C103	PH8151 - Engineering Physics	1.8	3	Attained
4.	C104	CY8151 - Engineering Chemistry	2.06	2.68	Attained
5.	C105	GE8151 - Problem Solving and Python Programming	1.83	2.57	Attained
6.	C106	GE8152 - Engineering Graphics	2.34	3	Attained
7.	C107	GE8161 - Problem Solving and Python Programming Laboratory	2.7	3	Attained
8.	C108	BS8161 - Physics and Chemistry Laboratory	2.7	3	Attained
2020-2021 Even Semester II Semester					
9.	C109	HS8251 - Technical English	2.16	2.9	Attained
10.	C110	MA8251 - Engineering Mathematics II	2.16	3	Attained
11.	C111	PH8201 - Physics of Civil Engineering	2.76	2.93	Attained
12.	C112	BE8251 - Basic Electrical and Electronics Engineering	1.8	2.71	Attained
13.	C113	GE8291 - Environmental Science and Engineering	2.14	2.71	Attained
14.	C114	GE8292 - Engineering Mechanics	1.8	2.92	Attained
15.	C115	GE8261 - Engineering Practices Laboratory	2.8	3	Attained
16.	C116	GE8211 - Computer Aided Building Drawing	2.76	3	Attained
2021-2022 Odd Semester III Semester					
17.	C201	MA8353 - Transforms and Partial Differential Equations	1.8	2.21	Attained
18.	C202	CE8301 - Strength of Material I	1.8	2.52	Attained
19.	C203	CE8302 - Fluid Mechanics	1.8	2.4	Attained
20.	C204	CE8351 - Surveying	1.9	3	Attained
21.	C205	CE8391 - Construction Materials	2.28	2.58	Attained
22.	C206	CE8392 - Engineering Geology	1.99	2.7	Attained

23.	C207	CE8311 - Construction Materials Laboratory	2.97	3	Attained
24.	C208	CE8361 - Surveying Laboratory	2.7	3	Attained
25.	C209	HS8381 - Interpersonal Skills/Listening and Speaking	2.4	3	Attained
2021-2022 Even Semester IV Semester					
26.	C210	MA8491 - Numerical Methods	1.8	2.12	Attained
27.	C211	CE8401 - Construction Techniques and Practices	2.18	2.4	Attained
28.	C212	CE8402 - Strength of Materials II	1.8	2.44	Attained
29.	C213	CE8403 - Applied Hydraulic Engineering	1.97	1.33	Not Attained
30.	C214	CE8404 - Concrete Technology	1.93	2.27	Attained
31.	C215	CE8491 - Soil Mechanics	1.8	2.34	Attained
32.	C216	CE8481 - Strength of Materials Laboratory	2.73	3	Attained
33.	C217	CE8461 - Hydraulic Engineering Laboratory	2.65	3	Attained
34.	C218	HS8461 - Advanced Reading and Writing	2.63	3	Attained
2022-2023 Odd Semester V Semester					
35.	C301	CE8501 - Design of Reinforced Cement concrete Elements	2	2.13	Attained
36.	C302	CE8502 - Structural Analysis I	2.13	2.46	Attained
37.	C303	EN8491 – Water Supply Engineering	2.29	2.13	Not Attained
38.	C304	CE8591 - Foundation Engineering	2.1	2.4	Attained
39.	C305	GE8071 - Disaster Management	2.41	2.47	Attained
40.	C306	ORO551 - Renewable Energy Sources	2.34	1.4	Not Attained
41.	C307	OAAN551 - Sensors and Transducers	1.8	0.2	Not Attained
42.	C308	CE8511 - Soil Mechanics Laboratory	2.88	3	Attained
43.	C309	CE8512 - Water and Waste Water Analysis Laboratory	2.88	3	Attained
2022-2023 Even Semester VI Semester					
44.	C311	CE8601 - Design of Steel Structures	2.53	2.58	Attained
45.	C312	CE8602 - Structural Analysis II	2.64	2.94	Attained
46.	C313	CE8603 - Irrigation Engineering	2.56	2.4	Not Attained
47.	C314	CE8604 - Highway Engineering	2.32	1.96	Not Attained
48.	C315	EN8592 - Wastewater Engineering	2.39	2.33	Not Attained
49.	C316	CE8611 - Highway Engineering Laboratory	2.97	3	Attained
50.	C317	CE8612 - Irrigation and	2.89	3	Attained

		Environmental Engineering Drawing			
51.	C318	HS8581 - Professional Communication	2.85	1.2	Not Attained
2023-2024 Odd Semester VII Semester					
52.	C401	CE8701 - Estimation, Costing and Valuation Engineering	2.4	2.4	Attained
53.	C402	CE8702 - Railways, Airports, Docks and Harbour Engineering	2.35	2.8	Attained
54.	C403	CE8703 - Structural Design and Drawing	2.09	2	Not Attained
55.	C404	EN8591 - Municipal Solid Waste Management	2.42	2.73	Attained
56.	C405	CE8711 -Creative and Innovative Project	3	3	Attained
57.	C406	CE8712- Industrial Training	2.98	3	Attained
2023-2024 Even Semester VIII Semester					
58.	C407	GE8076-Professional Ethics in Engineering (Elective IV)	2.43	1.36	Not Attained
59.	C408	CE8020- Maintenance, Repair and Rehabilitation of Structures (Elective V)	2.58	1.93	Not Attained
60.	C409	CE8811- Project Work	3	3	Attained

3.3. Attainment of Program Outcomes and Program Specific Outcomes (50)

3.3.1. Describe assessment tools and processes used for measuring the attainment of each of the Program Outcomes and Program Specific Outcomes (10)

For measuring the attainment of Program Outcomes and Program Specific Outcomes various tools are used. The process for measuring the attainment of each PO & PSO is described in Figure 3.3.

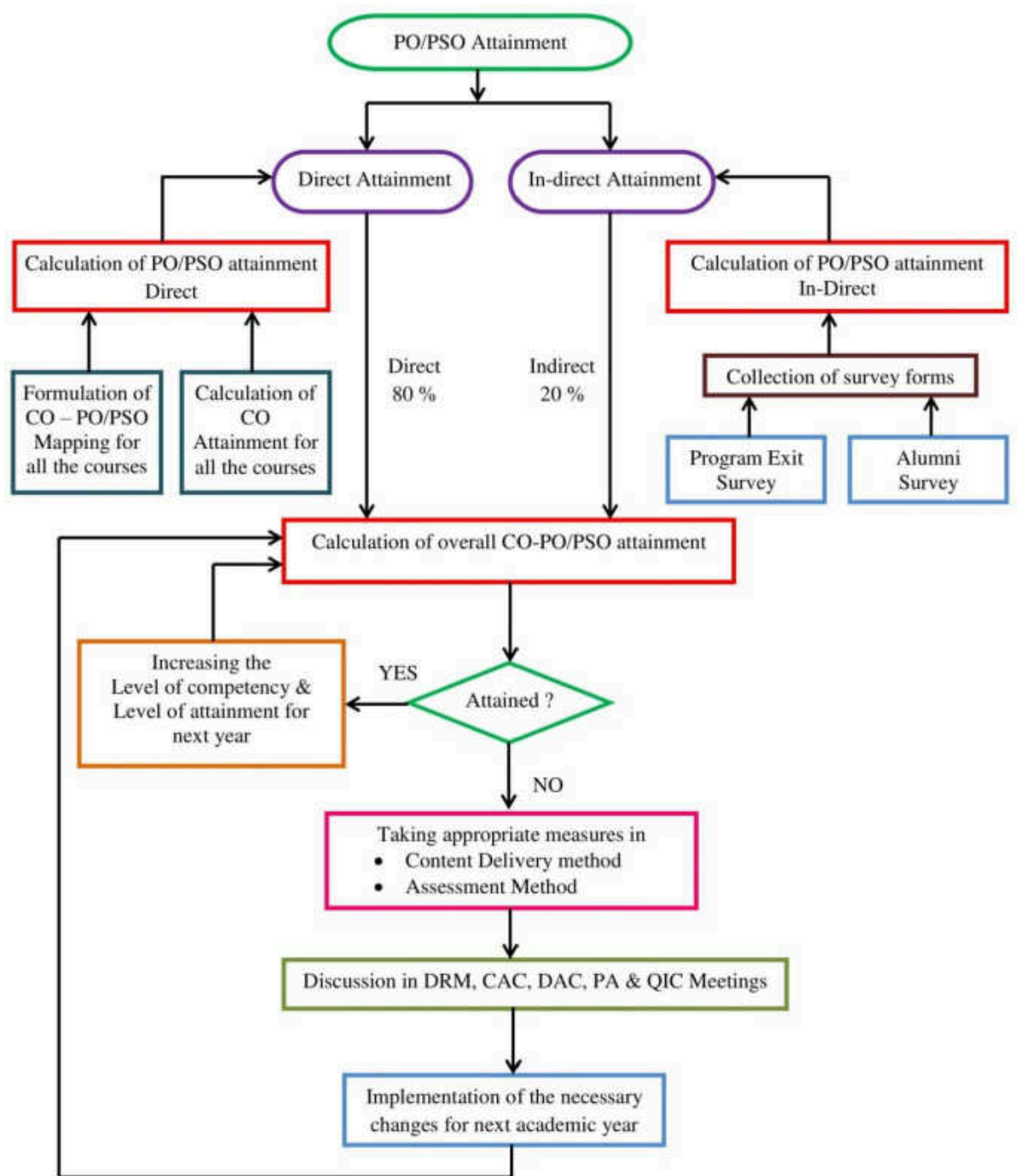


Figure 3.3 Process of PO & PSO Attainment

PO/PSO Assessment Tools:

Evaluation of attainment of POs and PSOs is based on direct and indirect assessment tools. Direct assessment of POs and PSOs is based on students' performance in Continuous Assessments and University Examination. Indirect assessment is based on Program Exit Survey, Alumni Survey and Course Outcome End Survey (Theory and Practical).

Direct Assessment:

Using Program Outcomes prescribed by NBA, the faculty member evaluates the Program Outcomes and Program Specific Outcomes through Internal Assessment Tests, Assignments / Tutorial, and Quiz. PO will be evaluated by the CO-PO Mapping with the attainment value for each course. For each course, every faculty member decides the competency level and attainment level.

The following table 3.3. shows the tools and process for direct PO attainment.

Table 3.3 Direct Assessment

PO Attainment	Tools	Process
Direct (CO Attainment)	- Internal Assessment Test - Assignments - Tutorials - Quiz - University Examination	- Assignments / Tutorials / online quizzes are given periodically for the entire course to attain the specific POs. - Three Internal Assessment Tests are conducted per semester to evaluate student performance. - University Examination is conducted once a semester as per the Anna University Schedule
	- Performance - Viva Voce - Record - Quiz	- Student Contribution in the laboratory is evaluated based on the performance, Viva Voce, Presentation, and Record Work. - Model Practical examination is conducted for 100 marks for 3 hours. - University Examination is conducted once a semester as per the Anna University Schedule
	Project Reviews	- Students are divided into batches. Each batch consists of two to four students. - Supervisors are allotted for each group. - Zeroth reviews are conducted for the students to identify the area of project. - Three reviews are conducted periodically to monitor and evaluate the progress of the project. - Viva-Voce is conducted at the end of the semester.

Course level PO & PSO Attainment Calculation:

The PO & PSO attainment for the course is calculated using the following formula.

$$CO \text{ Attainment Ratio of Course}(x) = \frac{CO \text{ Attainment of Course}(x)}{3(\text{Maximum attainment Value})}$$

Where, $x = [1 \text{ to } N]$, $N = \text{Number of Courses}$.

POm Attainment of course(x)

= CO Attainment Ratio of Course(x) X POm Mapping Value of Course(x)

Where, $m = [1 \text{ to } 12]$

PSOm Attainment of course(x)

= CO Attainment Ratio of Course(x) X PSOm Mapping Value of Course(x)

Where, $m = [1 \text{ to } M]$, $M = \text{Number of Program Specific Outcomes}$.

Program level PO & PSO Direct Attainment Calculation:

The PO & PSO attainment for Program is calculated using the following formula.

$$\text{POm Direct Attainment} = \frac{\sum_{i=1}^x \text{POm Attainment of course}(i)}{x}$$

Where, $m = \text{Program Outcomes varies from 1 to 12}$

$x = \text{Number of Courses mapped with POM}$

$$\text{PSOm Direct Attainment} = \frac{\sum_{i=1}^x \text{PSOm Attainment of course}(i)}{x}$$

Where, $m = \text{Program Specific Outcomes varies from 1 to 4}$

$x = \text{Number of Courses mapped with PSOM}$

Indirect Assessment:

The following tools are used to assess the indirect assessment of attainment of POs and PSOs. The assessment tools listed in table 3.4 are used for both PO – PSO attainment calculation.

Table 3.4 Tools for Assessment process

S. No.	Tools used for Assessment processes	Batch 2018-22	Batch 2019-23	Batch 2020-24
1	Program Exit Survey	✓	✓	✓
2	Alumni Survey	✓	✓	✓

Course Outcome End Survey (Theory & Practical):

The course outcome end survey is process of collecting reviews on each theory course and laboratory from the students at the end of each semester. It helps to improve the overall aspect of the course in future semesters. The survey covers the overall view about the learning outcome of the respective course/laboratory.

Program Exit Survey:

It is a process of collecting satisfaction survey on the quality of education from the perspective of graduating students upon the completion of their program. Program Exit Survey is structured with 5 like scale questions. The survey helps in identifying

- Perception on the overall quality of teaching, learning and mentoring.
- Opinion about the support provided by the programme in projects, modern tools and softwares.
- Support provided for extra-curricular and co-curricular activities.
- Exposure to the competitive exams and personality development programmes.
- Insight on imparting skills like entrepreneurship and societal responsibility through NSS, YRC, NGO and Club's
- Improvement on facilities.

PO & PSO Attainment Calculation of PES:***Question Level Calculation:***

$$Q(x) \text{ in } \% = \frac{\text{no. of Students provided more than or equal to 3 Marks in } Q(x)}{\text{no. of Students}} \times 100$$

Where, $x = [1 \text{ to } 25]$, $Q[x] = x^{\text{th}}$ Questions in Program Exit Survey

Question Level Attainment:

PES Attainment level		
Program Exit Survey (PES) Question Level Attainment	3	80% of the students provided more than or equal to 3 Marks in Survey Question
	2	70% of the students provided more than or equal to 3 Marks in Survey Question
	1	60% of the students provided more than or equal to 3 Marks in Survey Question

$$POm \text{ Attainment of PES} = \frac{\sum_{i=1}^x Q(x) \text{ Attainment mapped with } POm}{x}$$

Where, $m = [1 \text{ to } M]$, $M = \text{Number of Program Outcomes}$.

$x = \text{Number of Questions mapped with } POm$

$$PSOm \text{ Attainment of PES} = \frac{\sum_{i=1}^x PO(x) \text{ Attainment mapped with } PSOm}{x}$$

Where, $m = [1 \text{ to } M]$, $M = \text{Number of Program Specific Outcomes}$.

$x = \text{Number of POs mapped with } PSOm$

Alumni Survey:

The alumni survey is conducted through the survey questionnaire after graduation towards the achievement of POs and PSOs. Survey form is structured with six sections with respect to,

- Personal information.
- Employment/higher studies/entrepreneurship- details.
- Technical, professional, communication and general skills at present towards RIT contribution.
- Experience at RIT in projects, extra-curricular, co-curricular activities, personality development, sports, NSS and YRC facilities.
- Suggestions for further improvement.
- Suggestions for bridging curriculum gap and other valuable inputs.

PO & PSO Attainment Calculation of AS:

Question Level Calculation:

$$Q(x) \text{ in } \% = \frac{\text{no. of Students provided more than or equal to 3 Marks in } Q(x)}{\text{no. of Students}} \times 100$$

Where, $x = [1 \text{ to } 25]$, $Q[x] = x^{\text{th}}$ Questions in Alumni Survey

Question Level Attainment:

Alumni Survey Attainment level

Alumni Survey (AS) Question Level Attainment	3	80% of the students provided more than or equal to 3 Marks in Survey Question
	2	70% of the students provided more than or equal to 3 Marks in Survey Question
	1	60% of the students provided more than or equal to 3 Marks in Survey Question

$$POm \text{ Attainment of AS} = \frac{\sum_{i=1}^x Q(x) \text{ Attainment mapped with } POm}{x}$$

Where, $m = [1 \text{ to } M]$, $M = \text{Number of Program Outcomes}$.

$x = \text{Number of Questions mapped with } POm$

$$PSOm \text{ Attainment of AS} = \frac{\sum_{i=1}^x PO(x) \text{ Attainment mapped with } PSOm}{x}$$

Where, $m = [1 \text{ to } M]$, $M = \text{Number of Program Specific Outcomes}$.

$x = \text{Number of POs mapped with } PSOm$

Program level PO& PSO Indirect Attainment Calculation:

POm Indirect Attainment = (POm Attainment of PES X 0.5) + (POm Attainment of AS X 0.5) Where $m = \text{number of Program Outcomes}$

PSOm Indirect Attainment = (PSOm Attainment of PES X 0.5) + (PSOm Attainment of AS X 0.5)

Where $m = \text{number of Program Specific Outcomes}$

Overall PO & PSO Attainment Calculation:

The Overall PO & PSO attainment is calculated by using the following formula

Overall POm Attainment
= (POm Direct Attainment X 0.8) + (POm Indirect Attainment X 0.2)

Where $m = \text{number of Program Outcomes}$

Overall PSOm Attainment
= (PSOm Direct Attainment X 0.8) + (PSOm Indirect Attainment X 0.2)

Where $m = \text{number of Program Specific Outcomes}$

The following table 3.5. shows the tools and process for Indirect PO attainment.

Table 3.5 Tools for Assessment Process

PO Attainment	Tools	Process
Indirect	Program Exit Survey	On completion of the program, feedback is obtained from each student about the entire program experience through the Institute ERP portal
	Alumni Survey	During the alumni meet and graduation day the alumni survey is collected from the graduates based on the various parameters.

3.3.2. Provide results of evaluation of each PO & PSO (40)

PO Attainment

The representation of PO Attainment level in the following table as follows:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) ‘-‘ : No Attainment

BATCH 2019 - 2023

PO Attainment

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C101									2.20	2.80		
C102	2.64	2.29	1.94	0.88					0.88			1.23
C103	1.77	1.32	0.88	1.77	0.88	1.77					1.62	1.62
C104	1.94	1.06	0.88								2.64	2.64
C105	1.12	0.66	0.70	0.40	0.80							0.56
C106	2.46	2.64			1.76		1.76			2.64		0.88
C107	3.00	1.60	2.80		2.00							1.00
C108	2.66	3.00		3.00				2.66	2.33	2.00	3.00	3.00
C109									1.87	2.24		
C110	1.67	1.37	1.52		0.76							1.52
C111	2.24	1.44	0.80	2.24	0.80	2.24					1.76	1.76
C112	2.60	1.91	1.56	1.21								0.87
C113			2.00			2.00	3.00	2.00			2.00	3.00
C114	2.84	1.89	2.37	0.95	1.89						0.95	1.89

C115	1.83	1.00	1.00	1.00			2.83	2.00	2.00		2.00	
C116	3.00	2.00	1.00	3.00	3.00	2.00	1.00	3.00	1.00	2.00	2.00	2.00
C201	2.68	1.79	1.07	1.07								0.89
C202	2.76	1.84			0.92							
C203	3.00	2.40	1.60	1.75	1.00		2.00		1.00			1.00
C204	1.00	1.75				1.00						1.00
C205			2.84	2.84	0.95	2.84	0.95	1.89				
C206	2.53	1.42	2.84	1.26	1.42	1.26		0.95		0.95	0.95	0.95
C207	3.00	3.00							3.00			
C208	3.00			2.00	3.00				2.00			
C209									2.60	3.00		
C210	1.75	1.75	1.75	1.17	0.97						0.97	
C211	2.17	2.20	2.33	1.67	1.80	2.00	3.00	2.00		1.00		1.00
C212	3.00	3.00	2.00	1.00	3.00				1.00	2.00		2.00
C213	3.00	2.80	2.20	1.40	1.00							1.00
C214	3.00	2.00	2.00	1.50		1.30	1.20					1.00
C215	3.00	2.00	1.00	1.00		2.00	1.00	0.00	0.00	0.00	1.00	1.00
C216	3.00	2.00	1.00	3.00	3.00	2.00	1.00	1.00	1.00	1.00	2.00	2.00
C217	3.00	2.00	2.00	2.00								
C218									2.00	2.80		
C301	1.34	1.87	2.23	0.89	1.87							
C302	2.68	2.68	1.79	0.89	2.68				0.89	1.79		1.79
C303	3.00	2.00	3.00	2.00		2.30	2.00					2.00
C304	1.92	2.40	1.92	2.88						0.96		
C305	1.65	1.65	1.65			2.47			0.82	1.65		0.82
C306	1.87	0.93			1.87		1.87	0.93	2.15	1.87		0.93
C307	3.00	2.00	2.00	2.00	2.00	2.00		2.00	1.00			2.00
C308	3.00			2.00	3.00		1.50		2.00	2.00		
C309	3.00	2.00	2.00	2.00	3.00	3.00		2.00	3.00	2.00		2.00
C310	1.47	1.47	2.20	1.47	1.47	1.47	1.47	1.47	1.47	0.73	1.47	1.47
C311	2.76	2.76	1.84	0.92	2.76				0.92	1.84		1.84
C312	2.08	1.47	1.32		1.47	1.17	0.73	0.73	1.47			0.73

C313	1.77	2.12	1.98	1.06		1.77	0.71	1.59	1.41		1.41	
C314	2.73	1.82	2.73	1.82		0.91	1.82		0.91			1.82
C315	2.33	1.40	1.87			1.40	2.02					
C316	3.00			1.00					3.00			
C317	2.00	2.00	2.00				1.00			1.00		
C318									2.50	2.60		
C401	1.82	1.55			1.37	0.27					1.46	1.37
C402	2.34	1.76	1.47	0.98	1.95	0.98		0.98		0.98		0.98
C403	1.33	1.33	2.00	1.33	1.33	1.33	1.33	1.33	1.33	0.67	1.33	1.33
C404	1.95	0.98	1.95			1.95	1.95		1.95	1.22		1.47
C405	1.95	1.95	1.95	1.95			1.95					0.97
C406	3.00	3.00	3.00	3.00	3.00	2.00	2.00	1.00	3.00	3.00	2.00	2.00
C407	3.00	3.00	3.00	3.00	3.00	2.00	2.00	1.00	3.00	3.00	2.00	2.00
C408	1.56	1.34	0.89			2.67		2.67	2.67	2.67		0.89
C409	2.92	1.95	2.92				0.97	0.97	0.97			0.97
C410	3.00	3.00	3.00	3.00	2.00	2.00	2.00	1.00	3.00	3.00	2.00	2.00
Direct Attainment	2.49	1.93	1.67	1.64	1.61	1.86	1.77	1.72	1.63	1.87	1.74	1.47
Direct Attainment (80%)	1.992	1.544	1.336	1.312	1.288	1.488	1.416	1.376	1.304	1.496	1.392	1.176
Indirect Attainment	3	3	3	3	3	3	3	3	3	3	3	3
Indirect Attainment (20%)	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Total Attainment	2.592	2.144	1.936	1.912	1.888	2.088	2.016	1.976	1.904	2.096	1.992	1.776

2019-2023

PSO Attainment

Course	PSO1	PSO2	PSO3	PSO4
C101				
C102		0.88	1.32	
C103	0.88		1.77	0.88
C104	0.88			
C105		0.40	0.56	
C106	1.76			
C107	1.00	1.00	1.00	1.00
C108	1.66		1.00	1.00
C109				
C110		1.01	0.76	
C111	0.80		2.24	0.80
C112	1.91	1.73	0.87	
C113			1.00	
C114		0.95		
C115				
C116	2.00	3.00	2.00	3.00
C201			1.34	1.79
C202		0.92		
C203			1.80	1.40
C204		1.00	1.00	
C205		2.84	1.89	
C206	1.11	1.26	1.89	1.58
C207	2.80	3.00		3.00
C208		2.00	2.00	3.00
C209				
C210			1.22	
C211	1.50	2.00	1.00	1.33
C212	2.00	3.00	2.00	3.00

C213	1.50		1.60	2.00
C214	1.00	2.00	2.00	1.00
C215	1.00	1.00	1.20	1.33
C216	2.00	3.00	2.00	3.00
C217			3.00	
C218				
C301		1.42		2.52
C302	1.79	2.68	1.79	2.68
C303	1.00	3.00	2.75	2.50
C304	0.96	0.96	1.92	1.92
C305		0.82	0.82	
C306			0.93	
C307	2.00	3.00	2.00	3.00
C308	2.00	2.00	2.00	
C309	2.00	2.00	2.00	2.00
C310	1.47	2.20	1.47	2.20
C311	1.84	2.76	1.84	2.76
C312	0.73	1.10	1.47	1.22
C313	1.41	1.41	1.27	1.06
C314	0.91	2.73	2.50	2.28
C315		0.93		0.93
C316	2.80			3.00
C317	1.00		1.00	2.00
C318				
C401		1.37		
C402	1.47	0.98	1.47	0.98
C403	1.33	2.00	1.33	2.00
C404	1.47	1.47	1.79	
C405		1.62		
C406	3.00	2.00	2.00	2.00
C407	3.00	2.00	2.00	2.00
C408				1.78

C409	0.97	1.95	0.97	1.95
C410	3.00	2.00	2.00	2.00
Direct Attainment	1.49	1.72	1.52	1.82
Direct Attainment (80%)	1.192	1.376	1.216	1.456
Indirect Attainment	3	3	3	3
Indirect Attainment (20%)	0.6	0.6	0.6	0.6
Total Attainment	1.792	1.976	1.816	2.056